Notes: Madrian & Shea (QJE, 2001)

The Power of Suggestion: Inertia in 401(k) Participation and Savings Behavior

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1 Big picture

- **Question.** Does the default option in a retirement savings plan shape employee participation, contribution rates, and portfolio allocation?
- **Empirical setting.** A large U.S. company changed its 401(k) plan on April 1, 1998. New hires became automatically enrolled unless they opted out. The default was a 3% contribution rate invested entirely in a money market fund.
- **Core natural experiment.** Employees hired before and after the policy change faced different enrollment defaults but otherwise worked at the same firm and had similar plan features.
- **Main participation result.** Automatic enrollment increased 401(k) participation sharply. The raw comparison is 37.4% participation for the WINDOW cohort versus 85.9% for the NEW cohort; the regression-adjusted effect is about 50 percentage points.
- **Main contribution-rate result.** Automatic enrollment lowered average contribution rates among participants because many employees stayed at the 3% default. The regression-adjusted effect is about -2.2 percentage points.
- **Main asset-allocation result.** Automatic enrollment changed portfolios. Most NEW-cohort participants stayed in the default money market fund instead of allocating to stocks.
- **Default effect.** Among NEW-cohort workers, 61.1% both participated and stayed at the default contribution and fund allocation. Conditional on participation, 71.2% were at the default.
- **Immediate eligibility.** Earlier eligibility alone has a much smaller effect than automatic enrollment. It raises participation only modestly and has almost no effect on contribution rates.
- **Behavioral interpretation.** Defaults matter because of inertia, procrastination, complexity, and endorsement effects. The default is not just a fallback; it actively shapes saving decisions.
- **Policy takeaway.** Automatic enrollment can substantially increase plan participation, especially for low-income and short-tenure employees, but poorly chosen default contribution rates and asset allocations can also create conservative saving outcomes.

2 Incremental contribution of the paper

- **First major quasi-experimental evidence on automatic enrollment.** Before this paper, much of the discussion of 401(k) participation emphasized education, tax incentives, and individual choice. The paper provides clean evidence that changing the default option can change retirement saving behavior dramatically.
- **From participation to complete savings behavior.** The paper does not stop at enrollment. It studies contribution rates, asset allocation, and combined default behavior. This is important because automatic enrollment can raise participation while simultaneously lowering contribution rates and moving portfolios toward conservative assets.
- **Separates automatic enrollment from immediate eligibility.** The plan change combined automatic enrollment with immediate eligibility for new hires. By comparing OLD, WINDOW, and NEW cohorts, the paper shows that the big effect comes from automatic enrollment, not simply earlier eligibility.
- **Documents inertia in defaults, not just inertia in nonparticipation.** Earlier research often viewed inertia as failure to enroll. This paper shows that inertia also operates after enrollment: many employees participate but do not change the 3% money market default.
- **Links behavioral economics to retirement policy design.** The findings helped motivate the later policy focus on automatic enrollment, automatic escalation, and qualified default investment alternatives.

- **Clarifies a policy trade-off.** Defaults can be powerful welfare-improving tools, but the welfare effect depends on the quality of the default. A low contribution rate and a money market fund may increase participation but may not maximize long-run retirement wealth.

3 Data and measurement

3.1 Plan change and institutional setting

The firm changed its 401(k) plan on April 1, 1998. Before the change, employees became eligible after one year of employment and had to actively enroll. After the change, newly hired employees became eligible immediately and were automatically enrolled unless they opted out. The employer match still began only after one year of employment.

- The contribution range was 1% to 15% of compensation.
- The employer match was 50% of employee contributions up to 6% of compensation.
- Investment options remained unchanged: nine investment choices.
- The automatic-enrollment default was a 3% employee contribution invested 100% in the money market fund.

Interpretation: The plan change creates variation in the default rule while leaving many other plan features constant. This makes the setting powerful for studying default effects.

3.2 Cohorts used for comparison

The paper defines three key hire cohorts:

- **OLD cohort:** hired April 1, 1996 to March 31, 1997. These workers were eligible after one year and were not automatically enrolled.
- **WINDOW cohort:** hired April 1, 1997 to March 31, 1998. These workers became eligible around April 1, 1998 but were not automatically enrolled.
- **NEW cohort:** hired April 1, 1998 to March 31, 1999. These workers were eligible immediately and were automatically enrolled.

Interpretation: The WINDOW versus NEW comparison captures automatic enrollment plus immediate eligibility. The OLD versus WINDOW comparison isolates immediate eligibility because both cohorts are not automatically enrolled but differ in eligibility timing.

3.3 Data sources and observation dates

The analysis uses firm administrative data on employees and 401(k) plan records. The main observation date is June 30, 1999, about fifteen months after the automatic-enrollment change. The paper also uses June 30, 1998 and March 31, 2000 for some supplemental analyses.

The data include:

- 401(k) eligibility and participation;
- employee contribution rates;
- balances and contributions by fund category;
- employee demographics: age, gender, race/ethnicity, tenure, pay, work hours, and region.

Interpretation: Administrative records avoid many survey-measurement problems. The trade-off is that the paper studies one firm, so external validity must be interpreted carefully.

3.4 Main outcome variables

The main outcomes are:

$$\begin{aligned} Participation_i &= \mathbf{1}\{\text{employee } i \text{ participates in the 401(k)}\}, \\ ContributionRate_i &= \text{employee contribution as a percent of compensation}, \\ MoneyMarketShare_i &= \text{share of 401(k) balance or contributions in the money market fund.} \end{aligned}$$

The paper also measures whether an employee is at the automatic-enrollment default:

$$Default_i = \mathbf{1}\{Participation_i = 1,\ ContributionRate_i = 3\%,\ MoneyMarketShare_i = 100\%\}.$$

Interpretation: The default outcome combines three decisions. It captures whether the employee accepted the plan's entire suggested allocation rather than merely enrolling.

3.5 Fund categories

The paper aggregates fund choices into three broad categories:

- money market;
- stocks, including the foreign stock fund;
- bonds.

A balanced fund is split between stocks and bonds according to the fund's stated mix.

Interpretation: This aggregation makes the portfolio effects clear. The central comparison is not the choice among many funds but the contrast between the conservative money market default and the equity-heavy allocations chosen by non-default participants.

4 Models and empirical specifications

4.1 Raw cohort comparison

The simplest comparison is:

$$\Delta^{AE} = \bar{Y}_{NEW,6/30/99} - \bar{Y}_{WINDOW,6/30/98},$$

where Y is participation, contribution rate, or asset allocation. This comparison uses cohorts with similar tenure but different exposure to automatic enrollment.

For immediate eligibility, the analogous comparison is:

$$\Delta^{IE} = \bar{Y}_{WINDOW,6/30/99} - \bar{Y}_{OLD,6/30/98}.$$

Interpretation: The first comparison captures automatic enrollment, while the second captures earlier eligibility. The design uses cohorts whose tenure is made similar by choosing different observation dates.

4.2 Regression-adjusted effect

The paper also estimates regression-adjusted cohort differences:

$$Y_i = \alpha + \beta New_i + \gamma' X_i + \varepsilon_i,$$

where New_i indicates membership in the automatic-enrollment cohort and X_i includes categorical controls for gender, race, age, compensation, and months of tenure.

For immediate eligibility, the comparison replaces New_i with an indicator for the relevant eligibility cohort.

Interpretation: The regression adjustment asks whether cohort differences remain after controlling for observable demographic and tenure differences. In the paper, the adjusted automatic-enrollment effect is slightly larger than the raw effect for participation.

4.3 Default behavior decomposition

For the NEW cohort, the paper decomposes behavior into:

$$\Pr(Default) = \Pr(Participate) \times \Pr(Default \mid Participate).$$

It also defines a nondefault participation rate:

$$NondefaultParticipation = \Pr(Participate = 1, Default = 0).$$

Interpretation: This decomposition is important because a high participation rate can be driven by passive default enrollment. The nondefault participation rate measures active engagement with the plan.

4.4 Interpretation of treatment effects

The automatic-enrollment treatment changes two margins:

1. the participation default moves from nonparticipation to participation;
2. the default contribution and investment choices become 3% and money market.

Therefore, the effect on participation is expected to be positive, but the effect on contribution rates and risky asset shares may be negative.

Interpretation: The policy is not simply "more saving." It is a change in the choice architecture. The same intervention can increase the extensive margin of saving while lowering intensity and risk-taking among participants.

5 Identification and empirical design

5.1 Core identification strategy

The identification strategy exploits a firm-level policy change: automatic enrollment was introduced for workers hired on or after April 1, 1998. Employees hired just before and just after that date faced different defaults. The paper compares 401(k) outcomes across cohorts with similar tenure and demographics.

$$\text{Automatic enrollment exposure} = \mathbf{1}\{\text{hire date} \geq 4/1/1998\}.$$

Interpretation: The policy change is a natural experiment in choice architecture. The employer changed the default rule for new hires, and the authors study how behavior changes across cohorts.

5.2 Why the WINDOW versus NEW comparison identifies automatic enrollment

The WINDOW cohort and NEW cohort are compared at dates when they have similar tenure. The WINDOW cohort was hired in the year before the policy change and was not automatically enrolled. The NEW cohort was hired in the year after the policy change and was automatically enrolled.

The comparison is:

$$Y_{NEW,6/30/99} - Y_{WINDOW,6/30/98}.$$

Interpretation: This difference is designed to hold tenure roughly constant while changing the enrollment default. The large jump in participation from 37.4% to 85.9% is therefore interpreted as the effect of automatic enrollment.

5.3 Why the OLD versus WINDOW comparison identifies immediate eligibility

The plan change also introduced immediate eligibility for new hires. To avoid confounding immediate eligibility with automatic enrollment, the paper compares OLD and WINDOW cohorts. Both were hired before automatic enrollment, but they differ in when they became eligible.

The comparison is:

$$Y_{WINDOW,6/30/99} - Y_{OLD,6/30/98}.$$

Interpretation: The immediate-eligibility effect is much smaller than the automatic-enrollment effect. This supports the conclusion that default enrollment, not just earlier access, drives the main results.

5.4 Controls and regression adjustment

A concern is that the cohorts differ in observable characteristics such as age, pay, race, and tenure. The paper addresses this by using regression-adjusted comparisons with demographic controls:

$$Y_i = \alpha + \beta Treat_i + \gamma_1 Age_i + \gamma_2 Gender_i + \gamma_3 Race_i + \gamma_4 Compensation_i + \gamma_5 Tenure_i + u_i.$$

Interpretation: The control-adjusted estimates are close to the raw differences. This reduces concern that the results are merely driven by differences in cohort composition.

5.5 Timing and inertia

The analysis uses outcomes at multiple dates to study whether employees eventually undo the default. The March 31, 2000 analysis shows that many automatically enrolled employees remain at the default even after more time has passed.

Interpretation: If the default effect were only short-run confusion, it would disappear quickly. The persistence of default behavior supports an inertia or procrastination interpretation.

5.6 Threats to identification

- **Single-firm setting.** The setting is internally clean but may not generalize to all firms.

- **Cohort composition.** Workers hired at different times may differ. The paper addresses this with summary statistics and regression controls.
- **Combined plan changes.** Automatic enrollment coincided with immediate eligibility. The OLD-WINDOW comparison helps separate these channels.
- **Macroeconomic timing.** Cohorts are observed in different calendar years. Similar tenure comparisons and robustness checks reduce this concern, but cannot eliminate it completely.
- **Welfare ambiguity.** Higher participation may be beneficial, but a 3% contribution rate and money market allocation may be suboptimal for some workers.

Interpretation: The evidence is strongest for the behavioral fact that defaults matter. The welfare interpretation requires additional assumptions about optimal saving and portfolio choice.

6 Tables and figures

6.1 Table I: 401(k) Plan Features by Plan Date; Table II: Employee Cohorts for Comparative Analysis

Read:

- Table I shows that automatic enrollment was introduced after April 1, 1998, while many other plan features remained unchanged.
- Before the change, eligible employees had to actively choose participation; after the change, new hires were enrolled by default.
- The default contribution became 3% of compensation, and the default fund became the money market fund.
- Table II defines the OLD, WINDOW, and NEW cohorts and clarifies which cohorts experienced automatic enrollment and which experienced only earlier eligibility.

Economic message: These two tables establish the natural experiment. The default changes sharply for the NEW cohort, but employer match, vesting, loans, hardship withdrawals, and investment options remain broadly similar.

Detailed interpretation: The most important design feature is that automatic enrollment is not a vague encouragement. It changes the default action from nonparticipation to participation at a specific contribution rate and fund. This is why the later results can distinguish active savings decisions from passive acceptance of the employer's suggestion.

How it fits the paper: Table I gives the institutional treatment. Table II gives the comparison groups needed to identify that treatment.

TABLE I
401(k) PLAN FEATURES BY PLAN DATE

	Before 4/1/1998	After 4/1/1998
Eligibility		
Eligible employees	All except union and temporary employees	All except union and temporary employees
First eligible	After one year of employment	Immediately upon hire
Employer match eligible	After one year of employment	After one year of employment
Contributions		
Employee contributions	1 percent to 15 percent of compensation[a]	1 percent to 15 percent of compensation[a]
Employer match	50 percent of employee contribution up to 6 percent of compensation[a]	50 percent of employee contribution up to 6 percent of compensation[a]
Vesting		
Vesting of employee contributions	Immediate	Immediate
Vesting of employer contributions	2-year cliff	2-year cliff
Participation		
Default participation decision	No	Yes
Default contribution rate	None	3 percent of compensation
Default fund allocation	None	Money market fund
Other		
Loans	Available	Available
Hardship withdrawals	Available	Available
Investment choices	9 options	9 options

Source: Summary Plan Descriptions and personal communication with company officials.

Table I: 401(k) Plan Features by Plan Date. The key post-1998 changes are immediate eligibility and automatic enrollment with a 3% money market default.

TABLE II
EMPLOYEE COHORTS FOR COMPARATIVE ANALYSIS

	OLD	WINDOW	NEW
Dates of hire[a]	4/1/1996 to 3/31/1997	4/1/1997 to 3/31/1998	4/1/1998 to 3/31/1999
First eligible to participate in 401(k) plan	One year after date of hire	4/1/1998	Date of hire
First eligible for employer match	One year after date of hire	One year after date of hire	One year after date of hire
Automatically enrolled in 401(k) plan	No	No	Yes
Default contribution rate	None	None	3 percent
Default fund allocation	None	None	Money market fund

a. For employees hired through the acquisition of other companies, eligibility for 401(k) plan participation and the employer match was determined by the date of hire at the acquired company. Employees hired through acquisitions on or after 4/1/1998 were subject to automatic enrollment under the same terms as newly hired employees, with eligibility for the employer match determined by the date of hire at the acquired company.

Table II: Employee Cohorts for Comparative Analysis. The NEW cohort is exposed to automatic enrollment; the WINDOW and OLD cohorts are used to separate automatic enrollment from immediate eligibility.

6.2 Table III: Comparison of Worker Characteristics; Figure I: 401(k) Participation by Tenure

Read:

- Table III shows that the three main cohorts are broadly similar but not identical. The NEW cohort is younger and lower paid on average, consistent with shorter tenure and cohort turnover.
- Figure I shows that participation is usually strongly increasing in tenure before automatic enrollment.
- The NEW cohort has 86% participation despite low tenure, while the WINDOW cohort has 49% and the OLD cohort 57%.
- Longer-tenure pre-automatic-enrollment employees also have high participation, but they reach it only after many years.

Economic message: Automatic enrollment compresses years of participation accumulation into the first months of employment.

Detailed interpretation: The key visual contrast is that the NEW cohort looks like a high-tenure cohort even though it is newly hired. This is powerful evidence that defaults change behavior immediately. Table III matters because one must rule out the possibility that the NEW cohort is simply composed of workers more inclined to save. The demographic differences go the other way: the NEW cohort is younger and lower paid, groups that usually have lower participation.

How it fits the paper: Table III supports the validity of the cohort comparison; Figure I shows the behavioral magnitude of the default effect.

TABLE III
COMPARISON OF WORKER CHARACTERISTICS

	Study company				
	OLD cohort	WINDOW cohort	NEW cohort	All workers	U. S. workforce
Average age (years)	37.2	36.0	34.5	37.6	38.8
Gender					
Male	25.4%	23.9%	22.0%	22.1%	53.1%
Female	74.6	76.1	78.0	77.9	46.9
Ethnicity[a]					
White	77.1%	71.7%	68.8%	75.1%	74.6%
Black	12.5	16.8	18.9	14.1	11.3
Hispanic	7.1	8.2	6.7	6.6	9.5
Other	3.3	3.4	5.6	4.2	4.6
Hours					
Full-time (HPW > 35)	96.7%	95.6%	95.8%	94.6%	78.8%
Part-time (HPW < 35)	3.3	4.4	4.2	5.4	21.2
Compensation[b]					
Mean	$41,970	$38,424	$34,264	$40,180	$28,248
Median	$33,470	$30,530	$26,519	$31,333	$20,400
Geography					
East	17.0%	13.7%	16.9%	21.7%	18.9%
Midwest	38.2	34.9	31.0	32.8	24.1
South	28.2	33.0	32.0	31.3	34.7
West	14.6	16.1	19.6	13.1	22.4
Other[c]	2.0	2.3	0.6	1.1	—
Number of employees	N = 3286	N = 4257	N = 5812	N = 29,267	—

Authors' calculations. The sample in the first four columns is individuals employed at the study company on June 30, 1999. The sample in the last column is all individuals in the March 1998 Current Population Survey who worked in the previous year (weighted).

a. Ethnicity in the CPS is coded from the questions on race and Spanish ethnicity. We code individuals as follows: "White" if their race is white and they do not report Spanish ethnicity, "Hispanic" if their race is white and they do report Spanish ethnicity, "Black" if their race is black regardless of Spanish ethnicity, "Other" if their race is anything other than white or black regardless of Spanish ethnicity.

b. Compensation is the sum of annual base pay, incentive payments, and commissions.

c. The "Other" region is comprised of employees living and working in Puerto Rico and a few employees who live in Canada but work in the United States.

Table III: Comparison of Worker Characteristics. Cohorts differ somewhat by age and pay, motivating regression-adjusted estimates.

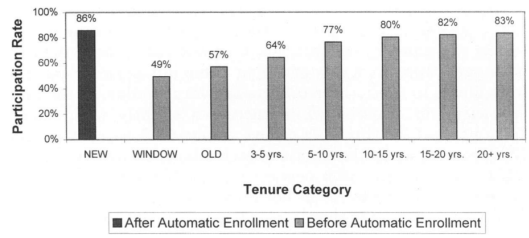

FIGURE I
401(k) Participation by Tenure

to 21 percent), and consequently compensation is higher at the

Figure I: 401(k) Participation by Tenure. The NEW cohort's participation is unusually high for a short-tenure group.

6.3 Table IV: Effects on 401(k) Participation; Table VI: Raw and Regression-Adjusted Effects

Read:

- Table IV reports 37.4% participation for the WINDOW cohort and 85.9% for the NEW cohort.
- Participation rises sharply for every demographic group under automatic enrollment.
- The low-income group below $20K increases from 12.5% to 79.5%, showing especially large gains among workers with low baseline participation.
- Table VI summarizes the effect: automatic enrollment increases participation by 48.5 percentage points raw and 50.4 percentage points regression-adjusted.
- Immediate eligibility has a much smaller participation effect: 0.6 percentage points raw and 4.1 percentage points adjusted.

Economic message: The participation result is large, broad-based, and not explained by demographics. Defaults are much more powerful than earlier access alone.

Detailed interpretation: The cross-group pattern is important. Automatic enrollment does not merely affect marginal high-income employees. It changes participation among groups historically less likely to save: younger, lower-paid, and minority employees. This is the part of the result most relevant for retirement policy, because automatic enrollment can reduce participation gaps.

How it fits the paper: Table IV provides the detailed heterogeneity, while Table VI extracts the treatment effect and shows that controls do not undo the result.

TABLE IV
THE EFFECTS OF AUTOMATIC ENROLLMENT AND IMMEDIATE ELIGIBILITY ON 401(k) PARTICIPATION

	Automatic enrollment		Immediate eligibility	
	Participation rate of Window cohort on 6/30/98	Participation rate of New cohort on 6/30/99	Participation rate of Old cohort on 6/30/98	Participation rate of Window cohort on 6/30/99
Overall	37.4%	85.9%	48.7%	49.4%
Gender				
Male	42.3	85.7	56.1	55.9
Female	35.9	86.0	46.3	47.4
Race/ethnicity				
White	42.7	88.2	53.4	54.4
Black	21.7	81.3	30.7	32.6
Hispanic	19.0	75.1	27.8	34.5
Other	46.2	85.2	55.0	62.9
Age				
Age <20	—	73.6	25.0	33.3
Age 20–29	25.3	82.7	36.7	36.9
Age 30–39	37.2	86.3	47.9	50.3
Age 40–49	47.3	90.1	54.9	58.0
Age 50–59	51.8	90.0	64.3	64.3
Age 60–64	60.0	86.0	60.6	70.0
Compensation				
<$20K	12.5	79.5	20.0	21.2
$20–$29K	24.5	82.8	31.7	35.3
$30–$39K	42.2	88.9	50.1	55.4
$40–$49K	51.0	91.8	61.6	64.5
$50–$59K	61.6	92.8	70.2	75.2
$60–$69K	59.7	94.7	79.2	75.1
$70–$79K	57.9	91.5	76.3	71.6
$80K+	68.3	94.2	76.3	82.6
Sample size	N = 4249	N = 5801	N = 3275	N = 4247

Author's calculations. The sample is 401(k) eligible active employees belonging to the cohort and

Table IV: Effects of Automatic Enrollment and Immediate Eligibility on 401(k) Participation. Automatic enrollment shifts participation upward across all demographic groups.

TABLE VI
RAW AND REGRESSION-ADJUSTED EFFECTS OF AUTOMATIC ENROLLMENT AND IMMEDIATE ELIGIBILITY

	Effect of Automatic enrollment: Window cohort on 6/30/98 vs. New cohort on 6/30/99	Effect of Immediate eligibility: Old cohort on 6/30/98 vs. Window cohort on 6/30/99
401(k) Participation rate		
Raw difference	48.5%*	0.6%
Regression-adjusted difference	50.4%*	4.1%*
401(k) Contribution rate		
Raw difference	−2.9%*	−0.1%
Regression-adjusted difference	−2.2%*	0.2%

Authors' calculations. The sample in the first two rows is 401(k) eligible employees belonging to the cohorts listed. The sample in the second two rows is 401(k) participants belonging to the cohorts listed. The sample is restricted to employees under the age of 65 as of the date listed in the column headings. The raw differences come from the first row of Tables V and VI. The regression-adjusted differences are estimated from OLS regressions which include categorical controls for gender, race, age, compensation, and months of

Table VI: Raw and Regression-Adjusted Effects. The adjusted participation effect of automatic enrollment is about 50 percentage points, while immediate eligibility is much smaller.

6.4 Figure IIa-IIc: Distribution of 401(k) Contribution Rates; Table V: Effects on 401(k) Contributions

Read:

- Figure II shows that the modal contribution rate under automatic enrollment is the 3% default.
- Among participants, 76% of the NEW cohort contributes at 3%, compared with about 12% of the WINDOW cohort.
- When nonparticipants are included, many WINDOW employees are at zero contributions, but NEW employees cluster at 3% rather than zero.
- Table V shows that the average contribution rate among participants falls from 7.3% for the WINDOW cohort to 4.4% for the NEW cohort.
- Table VI reports a -2.9 percentage point raw automatic-enrollment effect and a -2.2 percentage point adjusted effect on contribution rates.

Economic message: Automatic enrollment increases participation but anchors contribution rates at the default. This is the paper's central policy trade-off.

Detailed interpretation: The figure shows bunching, not merely a lower average. This matters because bunching at exactly 3% is hard to explain with standard preferences alone; it points directly to the default as a focal point. The contribution-rate effect also shows that participation effects can be misleading if the default rate is too low.

How it fits the paper: Figure II establishes the default anchor visually; Table V quantifies the average effect across demographic groups.

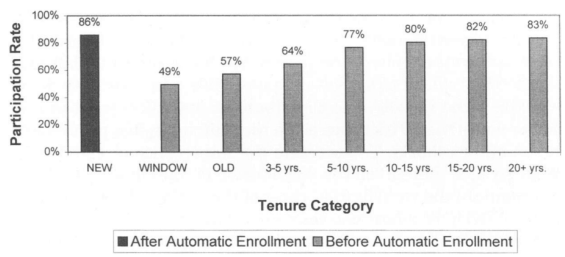

FIGURE IIa
Distribution of 401(k) Contribution Rates for 401(k) Participants

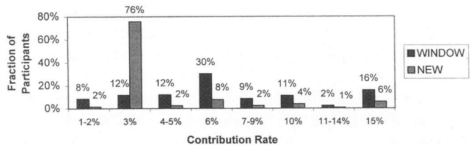

FIGURE IIb
Distribution of 401(k) Contribution Rates for the WINDOW and NEW Cohorts with Equivalent Tenure

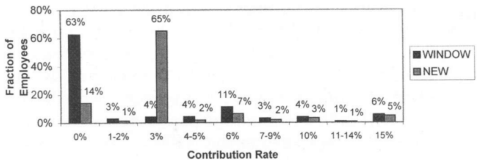

FIGURE IIc
Distribution of 401(k) Contribution Rates for the WINDOW and NEW Cohorts Including Nonparticipation

in the NEW cohort contributing at that rate (relative to around 10 percent of participants from the other cohorts).[9]

Figure IIa-IIc: Distribution of 401(k) Contribution Rates. The NEW cohort bunches at the 3% default, both among participants and when nonparticipants are included.

TABLE V
THE EFFECTS OF AUTOMATIC ENROLLMENT AND IMMEDIATE ELIGIBILITY ON 401(k) CONTRIBUTIONS

	Automatic enrollment		Immediate eligibility	
	Contribution rate of Window cohort on 6/30/98	Contribution rate of New cohort on 6/30/99	Contribution rate of Old cohort on 6/30/98	Contribution rate of Window cohort on 6/30/99
Overall	7.3%	4.4%	7.3%	7.2%
Gender				
Male	7.6	4.9	7.9	7.4
Female	7.1	4.2	7.0	7.1
Race/ethnicity				
White	7.5	4.7	7.4	7.5
Black	5.0	3.3	5.4	5.0
Hispanic	6.8	3.7	6.6	6.1
Other	8.9	5.0	9.2	8.8
Age				
Age <20	5.9	3.2	—	4.5
Age 20–29	7.1	3.8	6.4	6.0
Age 30–39	7.6	4.4	6.9	6.9
Age 40–49	8.8	4.9	7.8	7.6
Age 50–59	9.5	5.4	8.2	9.0
Age 60–64	6.0	6.9	9.5	9.7
Compensation				
<$20K	6.3	3.4	6.0	5.7
$20–$29K	5.5	3.5	5.9	5.7
$30–$39K	6.9	4.6	6.6	6.9
$40–$49K	7.7	5.2	7.8	7.7
$50–$59K	8.3	6.2	8.4	8.4
$60–$69K	8.7	7.1	8.6	9.1
$70–$79K	10.0	8.0	8.8	9.9
$80K+	8.8	6.6	7.7	8.2
Sample size	N = 1589	N = 4983	N = 1598	N = 2099

Authors' calculations. The sample is 401(k) participants belonging to the cohort and employed on the date

Table V: Effects of Automatic Enrollment and Immediate Eligibility on 401(k) Contributions. Automatic enrollment lowers average contribution rates among participants.

6.5 Table VII: 401(k) Asset Allocation and Contribution Statistics by Broad Fund Classification; Figure III: 401(k) Asset Allocation by Cohort

Read:

- Table VII shows that 90.8% of NEW-cohort employees with balances have some money market holdings, versus 17.0% for the WINDOW cohort and 18.2% for the OLD cohort.
- Among the NEW cohort, 75.3% have balances only in the money market fund.
- For contributions, 80.0% of NEW participants contribute only to the money market fund.

- Average contribution allocation in the NEW cohort is 80.5% money market, 16.4% stocks, and 3.1% bonds.
- Figure III summarizes the contrast: the 3+, OLD, and WINDOW cohorts allocate most contributions to stocks, while the NEW cohort allocates most contributions to the money market fund.

Economic message: The default affects not only whether employees save and how much they save, but also how they invest. Automatic enrollment can create conservative portfolios if the default fund is conservative.

Detailed interpretation: The NEW nondefault column is essential. NEW employees who actively move away from the default have much more stock exposure than the full NEW cohort. This shows that the money market allocation is not merely a cohort preference; it is tied to default acceptance. The asset allocation results therefore strengthen the inertia interpretation.

How it fits the paper: Table VII gives the detailed balance and contribution allocation statistics; Figure III makes the portfolio contrast visually immediate.

TABLE VII

401(k) Asset Allocation and Contribution Statistics by Broad Fund Classification

	Tenure/cohort					
	ALL	3+	OLD	WINDOW	NEW: All	NEW: Nondefault
Fraction with balances in fund type:						
Any money market	49.6%	42.0%	18.2%	17.0%	90.8%	71.3%
Any stocks	71.1	86.3	91.8	90.9	24.1	67.7
Any bonds	46.9	61.1	52.1	53.4	13.5	37.6
Fraction with balances in fund type:						
Only money market	24.3%	6.5%	5.2%	6.1%	75.3%	30.8%
Only stocks	22.3	24.1	39.5	38.9	5.0	15.5
Only bonds	2.2	3.3	2.3	2.0	0.2	0.5
Sample size	N = 19,935	N = 10,544	N = 1979	N = 2207	N = 5205	N = 1664
Fraction with contributions to fund type:						
Any money market	35.2%	19.3%	13.9%	13.9%	82.5%	39.3%
Any stocks	69.1	86.0	91.8	90.2	19.6	68.1
Any bonds	41.4	54.4	48.2	50.6	10.1	35.1
Fraction with contributions to fund type:						
Only money market	25.5%	5.1%	5.0%	6.4%	80.0%	30.6%
Only stocks	31.4	37.9	44.8	42.3	9.4	32.6
Only bonds	3.8	6.3	2.6	2.3	0.3	0.9
Average contribution allocation to fund type:						
Any money market	28.5%	9.8%	7.3%	8.2%	80.5%	32.5%
Any stocks	54.9	67.1	75.0	73.3	16.4	56.7
Any bonds	16.6	23.1	17.8	18.5	3.1	10.8
Sample size	N = 18,468	N = 9492	N = 1877	N = 2107	N = 4992	N = 1440

Authors' calculations. The sample for fund balance allocations is all active employees with nonzero 401(k) fund balances on 6/30/99. The sample for contribution allocations is all active 401(k) plan participants on 6/30/99. The sample is restricted to employees under the age of 65 and with three or more months of tenure.

Table VII: 401(k) Asset Allocation and Contribution Statistics by Broad Fund Classification. NEW-cohort participants are heavily concentrated in the default money market fund, while older cohorts are mostly invested in stocks.

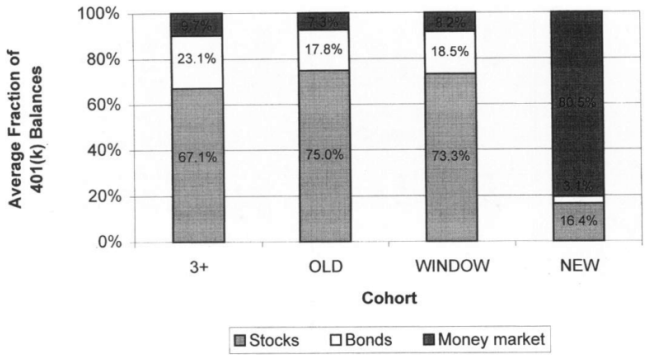

FIGURE III
401(k) Asset Allocation by Cohort

n of 401(k) contributions allocated to the various fund catego-

Figure III: 401(k) Asset Allocation by Cohort. The figure highlights the stark switch from stock-heavy allocations in pre-automatic-enrollment cohorts to money-market-heavy allocations in the NEW cohort.

6.6 Table VIII: "Default" 401(k) Participation and Investment Behavior; Figure IVa-IVc: Asset Allocation by Date of Initial 401(k) Participation

Read:

- Table VIII shows an overall default rate of 61.1% in the NEW cohort.
- Conditional on participation, the default rate is 71.2%.
- The nondefault participation rate is only 24.8%, meaning that active participation is much lower than the raw 85.9% participation rate.
- Default rates are especially high for low-income workers: 70.9% for those below $20K, and conditional default is 89.3%.
- Figure IV shows that employees who delayed initial participation until after the firm introduced automatic enrollment were also more likely to use the money market fund, even if they were not automatically enrolled.

Economic message: Much of the participation gain is passive acceptance of a bundle: participate, contribute 3%, and invest in the money market fund.

Detailed interpretation: Table VIII converts the main results into a single behavioral statistic. It shows that automatic enrollment is not simply a nudge into active saving; for many workers it becomes the entire savings plan. Figure IV is a subtle robustness and mechanism check: delayed participants in older cohorts also look more like default-takers, suggesting that procrastination and inertia are not unique to the NEW cohort but are made more visible by automatic enrollment.

How it fits the paper: Table VIII is the key evidence for default persistence; Figure IV connects default behavior to timing and delayed decision-making.

TABLE VIII
"DEFAULT" 401(k) PARTICIPATION AND INVESTMENT BEHAVIOR

	"Default" Rate		Nondefault 401(k) participation rate
	"Default" includes participation	"Default" conditional on participation	
Overall	61.1%	71.2%	24.8%
Gender			
Men	51.5	60.1	34.2
Women	63.8	74.3	22.1
Race/Ethnicity			
White	58.8	66.7	29.4
Black	71.2	87.8	10.0
Hispanic	57.4	76.5	17.7
Other	59.6	69.9	25.6
Age			
<20	66.7	90.6	6.9
20–29	64.3	77.8	18.3
30–39	60.9	70.6	25.3
40–49	59.2	65.7	30.9
50–59	53.7	59.7	36.6
60–64	37.2	43.2	48.8
Compensation			
<$20K	70.9	89.3	8.5
$20–$29K	69.0	83.3	13.8
$30–$39K	59.6	67.0	29.3
$40–$49K	51.5	56.1	40.4
$50–$59K	39.3	42.4	53.5
$60–$69K	38.5	40.1	56.3
$70–$79K	28.7	31.4	62.8
$80K+	34.2	36.3	60.0
Sample size	N = 5801	N = 4983	N = 5801

Authors' calculations. The sample is active 401(k) eligible employees belonging to the NEW cohort as of 6/30/99. The sample excludes employees who are aged 65 and over. In the first column, "Default" is defined as participating in the 401(k) plan at a 3 percent contribution rate invested 100 percent in the money market fund. In the second column the sample is restricted to 401(k) participants, and "Default" is defined as a 3 percent contribution rate invested 100 percent in the money market fund. In the third column the "Nondefault participation rate" is the fraction of employees participating in the 401(k) plan with a contribution rate other than 3 percent or an investment allocation other than 100 percent in the money market fund.

Table VIII: "Default" 401(k) Participation and Investment Behavior. Six out of ten NEW-cohort employees accept the full default bundle.

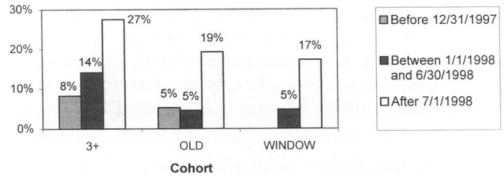

FIGURE IVa
Asset Allocation by Date of Initial 401(k) Participation: Average Money Marke Allocation

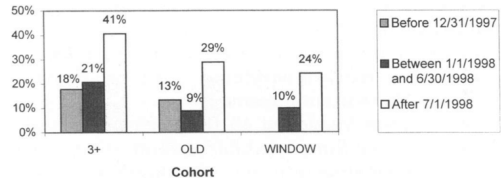

FIGURE IVb
Asset Allocation by Date of Initial 401(k) Participation: Any Money Market Contributions

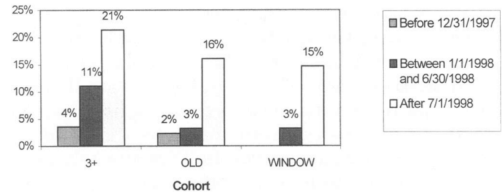

FIGURE IVc
Asset Allocation by Date of Initial 401(k) Participation: Only Money Market Contributions

Figure IVa-IVc: Asset Allocation by Date of Initial 401(k) Participation. Delayed participation is associated with more money market allocation, reinforcing the inertia interpretation.

7 Short summary

- The paper studies a large company's switch to automatic 401(k) enrollment for newly hired workers.
- The automatic-enrollment default is participation at a 3% contribution rate invested entirely in the money market fund.
- Participation rises dramatically: the NEW cohort has 85.9% participation versus 37.4% for the comparable WINDOW cohort.
- Regression-adjusted participation effects are about 50 percentage points.
- Earlier eligibility alone has only a small effect on participation.
- Automatic enrollment lowers average contribution rates among participants because many employees stay at 3%.
- Automatic enrollment shifts asset allocation toward the money market fund.
- 61.1% of NEW-cohort workers accept the full default bundle; 71.2% of participants do so.
- The effects are particularly important for low-income and younger workers, who historically have low participation.
- The paper shows that defaults can powerfully change saving behavior, but the default itself must be carefully designed.

8 Conclusion

8.1 Core conclusion

Madrian and Shea show that automatic enrollment has a large and immediate effect on retirement plan participation. The default option is not neutral. It changes behavior by moving employees from nonparticipation into participation without requiring an active decision.

8.2 Economic intuition

The economic intuition is that many employees face a complex savings decision with uncertain future benefits, paperwork costs, limited attention, and procrastination. When the default is nonparticipation, inertia keeps many workers out. When the default is participation, the same inertia keeps many workers in. The default acts as a suggestion, but also as a simplification of the decision.

8.3 Contribution-rate and portfolio implications

The paper also shows that automatic enrollment can have unintended effects. The 3% default contribution rate becomes a strong anchor, lowering average contribution rates among participants. The money market default similarly anchors asset allocation, producing very conservative portfolios for many newly enrolled workers.

8.4 Policy implications

The policy lesson is not simply that automatic enrollment is good. The lesson is that default design is crucial. A high-quality automatic enrollment policy should consider:

- a default contribution rate high enough to support retirement saving;
- automatic escalation to raise contributions over time;
- diversified default investments rather than a very conservative money market fund;
- easy opt-out and easy customization to respect heterogeneity in preferences and circumstances.

8.5 Limitations

The evidence comes from one company, so external validity is not automatic. The paper observes behavior but not the full welfare consequences for each worker. Some employees may rationally prefer low contributions or money market assets. Nevertheless, the clustering of behavior exactly at the default contribution and fund allocation is strong evidence that passive default acceptance is central.

8.6 Takeaway

The paper's lasting message is that the architecture of choice can be as important as the financial incentives themselves. Automatic enrollment changes participation, but default contribution and investment choices determine whether that participation produces adequate long-run savings.